A Multi-Model Machine Learning Approach for Predicting Success, Failure, and At-Risk Students' Status

Domain: Big Data Analytics in Education

6CS030 - Big Data

Proposal Presentation

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Problem Statement

The Engagement Gap in Online Learning

The Human Factor:

Unlike face-to-face settings, VLEs lack immediate physical cues, leading to:

The Data Paradox

Virtual Learning Environments (VLEs) generate huge amounts of data, yet dropout rates remain alarmingly high.

- **Social Isolation:** Lack of human connection reduces student persistence.
- **False Assumptions:** Students often withdraw early, mistakenly believing they are failing.

Solution:

A predictive system to identify at-risk students early enough for instructors to take preemptive action.

Research Question

How effectively can multi-model machine learning approaches (interaction, assessment and behavioral data) predict student success and at-risk status in the OULAD dataset?

Literature Base and Comparison

Base Paper and Comparison

Features	Base Paper: Students' Academic Performance and Engagement Prediction in a Virtual Learning Environment Using Random Forest with Data Balancing (Jawad et al., 2022)	Proposed Research
Model Focus	Single Model: **Random Forest**	**Multi-Model**: RF, k-NN, and Artificial Neural Networks (ANN)
Methodology	Profiles built on 6 time intervals (120–260 days)	**Feature Refinement**: Focus on high-impact behaviors (Forums/Clicks)
Data Handling	Utilized **Synthetic Data** generation	**Authentic Data**: No balancing; uses raw, natural distribution
Evaluation Metrics	Focus on predictive success vs. demographics	**Dual Metrics**: Accuracy + F1-score for Pass/Fail balance
Primary Goal	Proving VLE engagement > Demographics	Determining **Predictive Stability** across different algorithms

Supporting Paper and Relevancy

Study (Year)	Core Focus	Key Finding	Application to Proposal
Promoting Student Engagement in Online Education: Online Learning Experiences of Dutch University Students Vermeulen & Volman (2024)	**Behavioral**	Identifies the "why" and "how" behind engagement mechanisms (attention, effort, social interaction).	**Contextualization:** Allows for the categorization of VLE interactions (e.g., "social" forum clicks vs. "cognitive" content clicks).
How Well a Student Performed? A Machine Learning Approach to Classify Students' Performance on Virtual Learning Environment Alnassar et al. (2021)	**Algorithmic**	Compares k-NN, SVC, and ANN; finds k-NN is a top performer for engagement and performance indicators.	**Model Selection:** Guides the selection of k-NN and Artificial Neural Network (ANN) architectures for multi-model comparison.

Supporting Paper and Relevancy

Study (Year)	Core Focus	Key Finding	Application to Proposal
Dynamic Interaction between Student Learning Behaviour and Learning Environment Li & Xue (2023)	**Behavioral**	Isolates 14 influencing engagement factors via meta-analysis; identifies positive emotion as a critical internal predictor.	**Feature Engineering:** Translates raw clickstream data into meaningful behavioral and engagement features.
Impact of online learning on student's performance and engagement Akpen et al. (2024)	**Contextual**	Systematic review confirms that while performance can improve via online flexibility, engagement often declines due to isolation.	**Validation:** Validates the foundational premise of the predictive engine: using modeling to trigger interventions that solve the isolation gap.

Dataset and Source

Open University Learning Analytics dataset (OULAD)

Source: The Open University (UK) via the UCI Machine Learning Repository or Kaggle.

Link: OULAD Kaggle

Size: Approximately 465 MB (uncompressed), containing information on over 32,000 students.

Shape:
The dataset is **relational (structured)**,
consisting of **7** interconnected CSV files representing different aspects of the learning environment.

MetaData

.csv File	Core Information & Columns	Key Context for Analysis
courses.csv	code_module, code_presentation, length	Contains the list of modules and their starting periods (February "B" or October "J").
assessments.csv	code_module, code_presentation, id_assessment, assessment_type, date, weight	Details the types of assessments (TMA, CMA, Exam) and their weighted impact on the final grade.
vle.csv	id_site, code_module, code_presentation, activity_type, week_from, week_to	Catalog of available Virtual Learning Environment materials (HTML, PDF, etc.) and their planned usage weeks.
studentInfo.csv	code_module, code_presentation, id_student, gender, region, highest_education, imd_band, age_band, num_of_prev_attempts, studied_credits, disability, final_result	Information of the student, demographic, previous attempts, disabilities.
studentRegistration.csv	code_module, code_presentation, id_student, date_registration, date_unregistration	Tracks when students joined or withdrew from a module relative to the start date (day 0).
studentAssessment.csv	id_assessment, id_student, date_submitted, is_banked, score	Contains individual student results for specific assessments; scores below 40 are considered a Fail.
studentVle.csv	code_module, code_presentation, id_student, id_site, date, sum_click	**Big Data Interaction Log**; records the daily number of clicks a student makes on specific materials.

Aim

To develop and compare multiple Machine Learning models(Random Forest, k-NN, ANN) to accurately predict student sucess based on Virtual Learning Environment(VLE) engagement

Objectives

01

Data Structuring

Extract and merge demographic, assessment and clickstream features

02

Data Preprocessing

Cleanse dataset and perform Exploratory Data Analysis to find patterns

03

Feature Engineering

Segment the behavioural data across six progressive time intervals

04

Model Evaluation

Train models and bechmark multi-model performance using precision, recall and f1 metrics

THANK YOU.